

Data Gravity Map

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Compute follows where data lives

PROFESSIONAL DEFINITION

A Data Gravity Map illustrates how applications and compute are drawn toward where large data sets reside, because moving data is costly and slow — so 'data gravity' shapes architecture.

WHO DEVELOPED IT

Coined by Dave McCrory in 2010 to describe the pull that accumulated data exerts on services and infrastructure.

WHY IT WAS DEVELOPED

It was developed to explain why workloads cluster around data and to inform decisions about where to process AI workloads.

FIGURE 1 · DATA GRAVITY MAP

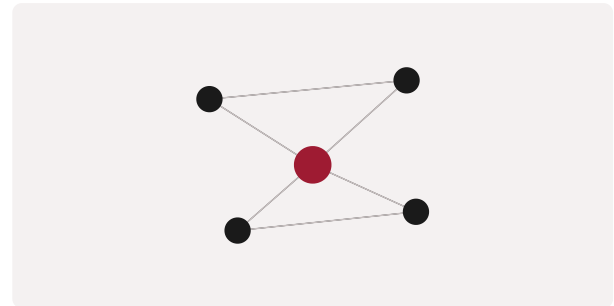


Figure 1. A schematic of the Data Gravity Map as applied in BARBM312 (illustrative).

HOW IT IS USED

Major data stores are mapped; compute and applications are located to minimise costly data movement, influencing cloud and edge choices.

WHEN IT IS USED

During data and AI architecture and multi-cloud planning.

BY WHOM IT IS USED

Architects, data engineers and infrastructure leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It names a force that quietly governs architecture and cost, guiding where AI processing should physically occur.

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SOURCES (HARVARD REFERENCING STYLE)

McCrory, D. (2010) 'Data gravity in the clouds', blog.

Cloud Security Alliance (2021) 'Data gravity and cloud architecture'.